

Baseball Mechanics AI — 12-Month Project Roadmap

Version 1.0 · 6 August 2026 · Planning horizon: Aug 2026 - Jul 2027

Team: 2 people, distributed (US / Korea). Combined capacity ~50 hrs/week (one full-time, one part-time).
Optimizing for: **a shippable product real athletes use.**

0. Executive Summary — The One Decision That Matters

The honest position is earlier than the repo history suggests. **One stage works end to end today: the 2D biomechanics overlay.** Pose extraction runs but is only partially reliable. Batter selection and swing segmentation are not working. 2D→3D lifting is not working. And event detection — the thing that makes every at-contact number trustworthy — was never built.

That means the first quarter is not "add event detection to a working pipeline." It is **restore a working pipeline, then add event detection.** Planning against the optimistic version of this baseline is the fastest way to lose the year.

So the strategic call for the next 12 months is narrow and unglamorous:

Ship batting only, to real users, with numbers you can defend. Treat pitching as a validated technical spike, not a shipped product. Do not touch golf or tennis.

The canvas vision — batting + pitching + tennis + golf, on iPhone + Android + web — is the right 5-year vision and the wrong 12-month plan. Two part-time people across a 13-hour time gap cannot ship four sports on three platforms. What they *can* do is ship one sport extremely well and build the architecture so that sport #2 costs 30% of what sport #1 cost. That compounding is the actual asset.

The 12-month definition of success:

#	Outcome	Measured by
1	Batting analysis in real athletes' hands	50+ athletes, 500+ analyzed swings, ≥40% return for a 2nd session
2	Numbers you can defend	Contact-frame timing MAE < 15 ms; key angles validated against OpenBiomechanics
3	Feedback that is cited, not hallucinated	100% of coaching statements linked to a metric ID + video timestamp; 0 banned claims
4	Unit economics that survive scale	< \$0.15 marginal cost per analyzed swing
5	Sport #2 is cheap	Pitching spike validated offline on shared infra; ≥70% code reuse demonstrated
6	The pipeline cannot silently break again	Stage-level regression tests green in CI; a deliberate break is caught automatically

1. Goal

Product goal

A single-camera phone app that turns a batting or pitching clip into body-mechanics data and coaching feedback, priced so a high-school athlete can afford it without a facility, a radar unit, or a wearable.

Why this can win

Every incumbent requires hardware or a building. Hawk-Eye needs a stadium. HitTrax and Rapsodo need purchased equipment. DiamondKinetics needs a bat sensor and still sees no body. The un-served market is the athlete with a phone and a parent who is not spending \$2,000. That is the entire youth and high-school market, and nobody is serving it credibly.

The non-negotiable principle

From your own vision document, and it should be framed on the wall:

Deterministic CV and geometry produce the measurements. The language model reviews, explains, and coaches. The language model never produces a number.

The moment an LLM is allowed to estimate bat speed from pixels, the product becomes unfalsifiable, un-reproducible, and — with minors — indefensible. This boundary is the product's integrity and it is not a technical preference, it is policy.

2. Where You Actually Are (Honest Baseline, Aug 2026)

Capability	State	Notes
2D biomechanics overlay	DONE Working	The one stage that works end to end. 4 angles drawn live; <code>2d_metrics.json</code> . This is the product's current floor.
Local batch pipeline (183 clips)	DONE Working	Streaming + subprocess batching, memory-bounded
Pose extraction (YOLO26m-pose + ByteTrack)	PARTIAL Partial	Runs, but not reliable enough to build on. Needs a measured failure profile before anything downstream can be trusted.
Training corpus	PARTIAL Partial	173 Penn Action swings + 10 pro slow-motion clips
Feedback engine (MEDA/APA/APE)	PARTIAL Conceptual	Roles defined; not running code
Batter selection + swing segmentation	NOT DONE Not working	Previously recorded as validated; it is not. Without it there is no reliable per-swing unit of analysis.
2D→3D lifting (MotionBERT)	NOT DONE Not working	Every view-independent metric is unavailable until this is restored.

Capability	State	Notes
Event detection	NOT DONE Never built	The gating dependency. Blocks every at-contact metric.
Bat tracking	DEFERRED Deferred / failing	Pretrained YOLO collapses to 17-18% in the contact zone
Capture quality gate	NOT DONE Missing	No protection against unusable uploads
App	NOT DONE Not started	—
Pitching	NOT DONE Not started	Taxonomy and phase model documented in canvas only

Reading of this table: roughly **25-30%** through the *measurement* problem and 0% through the *product* problem. Several stages previously recorded as complete are not. **Q1 is a recovery quarter, not an extension quarter.**

▲ **The documentation drift is itself a finding.** `MEMORY.md` records batter selection as "100% accurate across 444 frames" and 3D lifting as producing biomechanically sane angles (separation 55-61°). Neither holds today. Whatever the cause — regression, environment change, or a validation narrower than it appeared — **stage-level regression tests become a Q1 deliverable (M1.0).** A pipeline that can silently go from "validated" to "not working" without anyone noticing will do it again, and next time it may happen after you have users.

Check this before scoping anything: the 2026-06-12 decision record explicitly *decoupled* batter selection from bat detection — selection became largest-bbox-per-frame and `bat_boxes_raw.csv` was dropped as an input. If segmentation is currently blocked by failing bat tracking, that coupling may have been reintroduced by accident. If so, **decoupling it again is potentially a same-day unblock rather than a quarter of work.** Worth an hour of investigation before committing the quarter.

3. Physics Limits — Design Around These, Do Not Fight Them

These come from your own vision analysis and they cap what is honestly claimable. Every milestone below respects them.

Constraint	Consequence	Product response
30 fps = 33.3 ms/frame; a 70-90 mph bat travels 1.0-1.34 m between frames	Contact instant and peak bat speed are not recoverable at 30 fps	Capture gate rejects or downgrades 30 fps uploads for contact metrics
120 fps → 0.26-0.34 m/frame; 240 fps → 0.13-0.17 m/frame	Even 240 fps needs sub-frame fitting	120 fps = minimum for contact metrics; 240 fps recommended
Monocular 3D has ambiguous absolute scale and depth	"Metric" 3D from one phone is a claim you cannot support	Report angles (scale-free) confidently; gate distances/speeds behind calibration

Constraint	Consequence	Product response
2D wrist-path angle $\neq$ 3D bat-barrel attack angle	Current "attack angle" is a proxy, not the MLB metric	Rename in UI to " hand-path angle (proxy) " until bat tracking lands
Exit velocity requires ball trajectory + time + scale	Cannot be derived from the bat or body alone	Do not ship exit velocity in Year 1. Say so plainly.

Management note: the single highest-leverage non-code action in Q1 is writing a capture protocol that gets users to shoot at 120 fps+ on a tripod. A better input beats a better model, costs nothing, and no competitor can copy it away from you.

4. Technical Requirements — Batting

4.1 Athlete segmentation (drives the baseline comparison)

Per the canvas, three hitter archetypes, each with its own reference distribution:

Archetype	Profile	Reference athletes
Power	High exit velocity, higher strikeout rate	Judge, Ohtani, Raleigh, Schwarber, Alvarez, Caminero, Wood
Contact	High contact rate, low strikeout rate	Lee Jung-hoo, Ichiro, Arraez, Hoerner, Simpson, Clement
Gap / mid-power	Doubles, balanced profile	Olson, Rafaela, Burtleson, Devers, Freeman, Soto

Comparing a contact hitter against a power-hitter baseline produces confidently wrong advice. Archetype selection is a required user input, not optional.

4.2 Required user inputs

Archetype · height & weight · handedness (L/R) · batting video

4.3 Pipeline requirements

#	Layer	Requirement	Status
1	Capture gate	Score resolution, true FPS, shutter blur, camera motion, full-body visibility, view angle, occlusion, lighting. Reject early, before expensive inference.	Build Q1
2	Pose	PARTIAL Partially working — stabilize first. Measure the failure profile (which joints, which views, which frame rates) before extending. COCO-17 is insufficient for hands/pelvis/thorax; extend only after it is reliable.	Fix Q1 / extend Q2
3	Batter selection + segmentation	NOT DONE Not working — restore first. Largest-bbox-per-frame selection, <code>segment_id</code> on track change. Verify it is not incorrectly coupled to failing bat detection.	Fix Q1 — first task

#	Layer	Requirement	Status
4	Player isolation	Instance segmentation mask to remove background people; batter identity score from uniform + box ROI + temporal keypoint quality, not ByteTrack ID alone.	Q1
5	Keypoint quality	Per-keypoint confidence/covariance; explicitly tag <code>observed</code> / <code>interpolated</code> / <code>proxy</code> . Never blend them silently.	Q1
6	3D lifting	NOT DONE Not working — restore. MotionBERT behind a swappable adapter. Separate camera-space / world-space / image-space by type. Mirror transform for L/R in exactly one place. If it cannot be restored in 3 weeks, ship 2D-only and defer 3D to Q3 (see fallback below).	Fix Q1
7	Event detection	8-state model: address → load → stride → foot plant → launch → contact → extension → finish. Inputs: body keypoints, bat axis, ball trajectory, audio impact transient. Every event carries a timestamp + confidence interval.	Q1-Q2 — top new build
7	Metrics	Hip-shoulder separation, spine tilt, front-knee angle, lead-arm extension, hand-path angle (proxy), time to contact. Each carries value, unit, space, event, confidence, CI, source joints/frames, definition version.	Q2
8	Bat tracking	TrackNet-style temporal model on knob + tip, motion-blur augmentation, ~200-500 labeled frames including contact-zone frames .	Q3 (conditional)
9	Blocked metrics	Bat speed (needs scale calibration). Exit velocity (needs ball tracking). Do not ship in Year 1.	Year 2

4.4 Audio is underrated

Bat-ball contact produces a sharp audio transient that a phone mic captures at 44.1 kHz — roughly **1,470× the temporal resolution of a 30 fps video frame**. Audio will not tell you *where* contact happened, but it pins *when* to within a millisecond. For a project whose central weakness is contact timing, this is the cheapest accuracy win available. Prototype it in Q1 alongside the visual event model.

5. Technical Requirements — Pitching

Pitching reuses the skeleton backbone but is **not** a configuration change. Three things are genuinely different: the event model, the arm-slot taxonomy, and the risk profile.

5.1 Arm-slot segmentation

Slot	Description	Reference pitchers
Overhand	Near-vertical delivery, downward plane	Schlittler, Cole, Yesavage, Chapman, Miller
Three-quarter	Between overhand and sidearm; most common	Ohtani, Woo, Castillo, Yamamoto
Sidearm	Near-horizontal arm path	—

Slot	Description	Reference pitchers
Underhand / submarine	Low-to-high, rising path	—

Arm slot should be **auto-classified from the video and confirmed by the user**, not typed in blind — most amateur pitchers misreport their own slot.

5.2 Kinetic chain (the thing being measured)

Ground → legs → pelvis → trunk → shoulder → elbow → hand → release.

Pitching analysis is fundamentally about **sequencing and timing**, not static positions. The question is never "is the elbow in the right place" but "did the pelvis start rotating before the trunk, and by how many milliseconds." This makes event timing even more central than in batting.

5.3 The 9 events

first movement · peak leg lift · hand separation · max stride / foot hover · **lead-foot contact** · **maximum external rotation (MER)** · **ball release** · max lead-knee extension · finish

Two of these are hard from a single phone camera and must be flagged as such:

- **MER** — peak shoulder external rotation happens in a few milliseconds with severe forearm motion blur. Below 240 fps, treat as unmeasurable.
- **Ball release** — needs either ball detection or a very high frame rate. Without it, release-point consistency (arguably the single most useful pitching metric) is not available.

5.4 Multi-pitch requirement — a product constraint, not a technical one

Your own canvas states it: judge across several pitches, never one delivery. A single-pitch pitching report is professionally misleading. **The pitching product must require a minimum of 5 pitches per session** and report variance, not just the mean. Release-point *consistency* is the deliverable; release point is only an input.

5.5 Safety policy — decide this before writing pitching code

You will be analyzing throwing mechanics for minors. Pitching mechanics are causally linked in public discourse to elbow and shoulder injury. Therefore:

The product must never output injury risk, injury prediction, or medical language — not as a hedge, not as a percentage, not as a "flag." No ground truth exists in your data to support such a claim, and the population is children.

Ship descriptive mechanics ("your trunk begins rotating 40 ms before lead-foot contact, earlier than your archetype baseline") and never diagnostic claims. Encode this as a banned- phrase list enforced in the output schema validator, not as a guideline anyone has to remember.

5.6 Shared vs. sport-specific — the reuse thesis

Reusable across sports (~70%)	Sport-specific (~30%)
Capture quality gate	Event state model

Reusable across sports (~70%)	Sport-specific (~30%)
Pose extraction + tracking	Metric definitions
Player isolation + identity	Archetype taxonomy
3D lifting adapter	Baseline reference library
Evidence Packet schema	Coaching rubric
Feedback engine (MEDA/APA/APE)	Capture protocol specifics
App shell, auth, storage, billing	—

This table is the business case for the multi-sport vision. Golf and tennis are, structurally, "new event model + new metric definitions + new baseline library" on top of an existing spine. Prove that with pitching in Q4 and the expansion story becomes credible to a partner, an investor, or a hire.

6. Technical Suggestion — The AI Agent / Multimodal Architecture

6.1 The architectural boundary

DETERMINISTIC LAYER — produces every number
capture gate → pose → tracking → events → 3D → metrics
Output: Evidence Packet (JSON, schema-validated)

numbers cross this line ONE WAY

REASONING LAYER — explains, never measures
MEDA → APA → APE, backed by local VLM lanes
Output: coaching report, every claim citing an evidence_id

6.2 The Evidence Packet — the single most important interface

Everything the reasoning layer sees, it sees through this contract. Nothing else.

```
{
  "run_id": "2026-08-06-athlete042-swing03",
  "athlete_profile": { "bats": "R", "age_band": "HS", "archetype": "contact",
    "height_cm": 178, "weight_kg": 74 },
  "capture_quality": { "fps": 240, "view": "side", "score": 0.91,
    "flags": ["contact_blur"] },
  "events": [ { "name": "foot_plant", "t_ms": 610, "confidence": 0.88 },
    { "name": "contact", "t_ms": 842, "confidence": 0.76 } ],
  "metrics": [ { "id": "m_hss_01", "name": "hip_shoulder_separation",
    "value": 31.2, "unit": "deg", "space": "world",
    "event": "launch", "ci95": [27.8, 34.9],
    "confidence": 0.83, "definition_version": "1.2",
    "source_joints": ["L_hip", "R_hip", "L_shoulder", "R_shoulder"] } ],
  "baseline_comparison": [ { "metric_id": "m_hss_01", "cohort": "contact_HS",
    "cohort_p50": 38.4, "cohort_iqr": [32.1, 44.0],
    "percentile": 24 } ],
  "keyframes": ["address", "foot_plant", "launch", "contact", "finish"],
  "overlays": ["pose", "hand_path", "uncertainty"]
}
```

Design rules:

1. If a metric is not in the packet, the agent cannot discuss it. This alone eliminates most hallucination.
2. Every metric carries a CI. The agent must soften or abstain when CI is wide.
3. `baseline_comparison` is computed deterministically by retrieval — **not** by the LLM.
4. Schema-validate on the way in and on the way out. Retry on failure, abstain after 2 retries.

6.3 Two-lane local VLM

Per your hardware analysis (M5 Max, 40-core GPU, 128 GB unified memory):

Lane	Model	Size	Role
Fast	<code>google/gemma-4-12B-it-qat-q4_0-gguf</code>	~6.98 GB	Capture-quality QA, keyframe description, draft reports, UI responses
Strong	<code>mlx-community/Qwen3.6-35B-A3B-4bit</code>	~20.4 GB	Escalation only: low confidence, L/R confusion, multi-swing comparison, final coaching synthesis

Both fit in memory; do not co-resident them at first. Keep the fast lane resident, load the strong lane on demand. **Escalation should be < 20% of runs** — if it exceeds that, the deterministic layer is under-performing and that is where the fix belongs, not in a bigger model.

Do not accept vendor benchmarks. Confirm lane assignment on your own golden set, measuring: Korean and English coaching accuracy, frame ordering, left/right mirroring, temporal causality, JSON schema compliance, and hallucination rate.

6.4 The three agents as running code

Today MEDA/APA/APE are conceptual roles. Make them services with typed contracts:

Agent	Input	Output	Implementation note
MEDA — target definition	Athlete profile (archetype, height, weight, handedness, age band)	Target mechanics envelope + rationale	Mostly retrieval, not generation. Query the versioned baseline library for the matching cohort. The LLM writes prose around retrieved numbers; it must never invent the numbers.
APA — current-state analysis	Evidence Packet	Pros/cons by mechanical category, each citing metric IDs	Deterministic comparison first, LLM narrates second.
APE — recommendations	MEDA target + APA analysis	Ranked, athlete-specific corrections + drills	Must state <i>why</i> each correction follows from this athlete's specific gap. Generic advice = a failed run.

Guardrails, enforced in code:

- Output schema requires an `evidence_id` on every claim; unparseable output is rejected, not patched.
- Banned-claim list (injury risk, medical language, "ideal," certainty language) checked post-generation.
- Abstain path: if capture score or metric confidence is below threshold, return "re-shoot" guidance instead of analysis. **An honest abstention is a feature.** It is also the cheapest possible response.

6.5 Baseline Reference Library — the piece that makes feedback credible

The canvas promises feedback derived from professional samples. Implement that as a **versioned metric distribution store**, not video similarity search:

```
baseline_library/  
  v1.2/  
    batting/  
      power_HS.json      # p10/p25/p50/p75/p90 per metric, n, source, capture conditions  
      contact_HS.json  
      gap_HS.json  
      power_pro.json     # aspirational reference tier  
    pitching/  
      overhand_HS.json  
      three_quarter_HS.json
```

Comparison is then a percentile lookup — fast, explainable, testable, and versioned so a baseline change never silently rewrites past reports. Store cohort `n` and surface it: a cohort of 6 athletes must not be presented with the authority of a cohort of 300.

6.6 Cost architecture

Local-first inference on the M5 Max is the right call for development and for the pro tier. For consumer scale, the economics that matter:

- Deterministic CV is the expensive part; VLM reasoning over a small JSON packet is cheap.
- Cache aggressively — the same athlete re-analyzed should not re-run pose extraction.
- The capture gate is a **cost control**, not just a quality control: rejecting a bad upload in 200 ms instead of running a full pipeline is the single largest lever on unit economics.
- Target < \$0.15 per analyzed swing. Instrument this from Q2, not Q4.

7. Video Collection Strategy — Building Local Model Baselines

You need three distinct things, and conflating them is a common and expensive mistake:

Need	Purpose	Volume	Quality bar
A. Training data	Fine-tune pose/event/bat models	Thousands of frames	Varied, realistic, messy
B. Ground truth	Validate metric accuracy	Hundreds of trials	Lab-grade, instrumented
C. Baseline library	Cohort comparison distributions	30+ per cohort	Consistent capture protocol

Tier 1 — Public datasets (free, start here)

Source	Contents	Use
Driveline OpenBiomechanics	100 pitchers + 98 hitters , C3D marker data: 47 body markers (22 lower-limb/pelvis, 25 head/upper-limb/trunk) plus 10 bat markers . Free for educational use through 31 Dec 2030.	This is your ground truth (Need B). The single most valuable external asset available to you. Validate hip-shoulder separation, sequencing, and lead-arm extension against real marker-based mocap.
Penn Action	2,326 clips / 15 actions; 173 baseball swing sequences already extracted to Penn_Action/batting/ , 7,561 frames, 13 joints + bbox + visibility	Pose/event pre-training and pipeline regression (Need A). Note: low resolution, ~30 fps, uncontrolled — good for robustness, useless for contact metrics.
MLB-YouTube (piergiaj/mlb-youtube)	Broadcast segments, activity-labeled	Raw video source only; needs pseudo-labeling + manual correction
SportsPose / AthleticsPose	3D sports pose datasets	Cross-domain 3D lifting validation

Priority action: license-check and download OpenBiomechanics in Week 1. Everything in your validation gate depends on having real ground truth, and you currently have none.

Tier 2 — Your own capture (highest value per unit effort)

Public data will never match your capture conditions. 30–50 athletes shot to your own protocol beats 5,000 scraped broadcast clips for baseline purposes.

Capture protocol (write this in Q1, it is a deliverable):

- Fixed tripod, 1080p minimum, **120 fps minimum / 240 fps preferred**
- Two views where possible: side (sagittal) and front (frontal)
- A known-length reference object in frame (a bat of known length works) for scale
- Preserve original timestamps; when converting variable to constant frame rate, keep a **time mapping table**, never bare frame indices
- 5+ swings or pitches per session
- Consistent lighting, clean background, full body visible throughout

Sourcing, in ascending order of effort:

1. Yourself and friends — start this week, zero coordination cost
2. Local high-school teams and academies — offer free analysis in exchange for footage rights
3. Korea-side advantage: your partner can run a parallel cohort in a different baseball culture. **Do not treat this as redundancy — treat it as your generalization test.** A model that works in both the US and Korea is meaningfully more robust than one tuned to a single region, and this is a real technical benefit of a distributed team.
4. Youth showcases and tournaments — high volume, but consent overhead is real

Consent and minors: get written parental consent before any footage of a minor is captured, stored, or used for training. Build the consent form in Q1 alongside the capture protocol, not after you have 200 clips you cannot legally use.

Tier 3 — Partnership (Q3+)

A single relationship with a facility that runs Rapsodo, HitTrax, or Blast Motion gives you paired video + instrument readings — exactly the ground truth needed to validate bat speed and exit velocity later. Trade free analysis for data rights. One good facility partnership is worth more than a year of scraping.

Labeling

- **Tool:** CVAT (already in your stack)
- **Event labels:** 200–500 swings labeled with the 8 batting events; this is the training set for Q1's event detector and it is on the critical path
- **Bat knob/tip:** 200–500 frames **including contact-zone blur** — bootstrap from YOLO boxes, hand-correct the blurred frames
- **Practical note:** event labeling is ~2–4 min per swing. 300 swings ≈ 15–20 hours. This is well-suited to the part-time partner, is fully async, and needs no shared working hours.

Legal caution

Broadcast footage (MLB, YouTube) is fine for internal research and model development. Redistributing it, or showing it inside a commercial product as reference material, is not. Keep two clearly separated buckets — `research_only/` and `commercial_ok/` — from day one. Retrofitting that separation later is painful.

8. Major Milestones & Timeline

Q1 · Months 1-3 (Aug-Oct 2026) — "Restore the Foundation"

Theme: ▲ Recovery quarter. Three broken stages must work again before anything new is built on top of them. No product work. Event detection **starts** here but is expected to finish in Q2 — that is the cost of the revised baseline, and pretending otherwise would just move the failure later.

Sequencing matters: do these in order. Each one gates the next.

Milestone	Deliverable	Owner
M1.0 Regression harness	Per-stage smoke tests + a 10-clip fixture set that fails loudly. Build this first — it is why the drift went unnoticed.	FT
M1.1 Batter selection + segmentation restored	Largest-bbox selection + <code>segment_id</code> working again. First check whether it is wrongly coupled to failing bat detection — that may be a same-day fix.	FT
M1.2 Pose reliability profile	Quantify where pose fails (joint, view, fps, occlusion). Stabilize to a known-good baseline. No extension work yet.	FT
M1.3 3D lifting restored	MotionBERT producing sane angles again, behind a swappable adapter. 3-week timebox — then take the fallback.	FT
M1.4 Ground truth acquired	OpenBiomechanics downloaded, license-verified, loaded, mapped to your joint schema	PT
M1.5 Capture protocol + consent	Written protocol, example videos, parental consent form	PT
M1.6 Golden set labeled	200–500 swings, 8 events each, in CVAT	PT
M1.7 Capture quality gate	Scores 8 dimensions, rejects before inference	FT
M1.8 Metric schema v1	Every metric carries value/unit/space/event/CI/version	FT
M1.9 Event detection started	Audio transient prototype + visual state model v0. Completion target moves to Q2.	FT

Q1 exit gate — all five: 1. Regression harness green, and it demonstrably catches a deliberately introduced break 2. Batter selection + segmentation working on the full 183-clip corpus 3. Pose failure profile documented with a known-good operating envelope 4. 3D lifting restored **or** the 2D-only fallback formally taken 5. Capture gate correctly rejects known-bad uploads

FALLBACK: Fallback if 3D cannot be restored in 3 weeks: ship the 2D-only product. Your 2D overlay works today, and hip-shoulder separation, spine tilt, and front-knee angle are genuinely coachable from 2D alone with a stated view-dependence caveat. 3D moves to Q3 as an accuracy upgrade. This is a perfectly good product — do not let a broken 3D stage hold the whole year hostage. Most competitors at this price point offer nothing comparable.

Q2 · Months 4-6 (Nov 2026-Jan 2027) — "Trustworthy Feedback"

Theme: finish event detection, then turn numbers into coaching. First real users.

Milestone	Deliverable	Owner
M2.0 Event detection shipped	<code>events.json</code> with per-event timestamp + CI; contact MAE < 15 ms at 240 fps . Carried over from Q1.	FT
M2.1 Evidence Packet v1	Schema, validator, generator wired to the pipeline	FT
M2.2 Baseline library v1	3 batting archetypes × HS tier, from own + public capture	PT

Milestone	Deliverable	Owner
M2.3 Fast lane live	Gemma 4 12B running locally; capture QA + draft reports	FT
M2.4 MEDA/APA/APE as services	Typed I/O, schema validation, banned-claim enforcement, abstain path	FT
M2.5 Validation report	Your metrics vs. OpenBiomechanics; unsupportable claims removed from all copy	Both
M2.6 Closed alpha	10 real athletes , full loop, structured feedback collected	Both
M2.7 Cost instrumentation	Per-swing cost tracked from day one	FT

Q2 exit gate: contact MAE < 15 ms; 10 athletes have received reports; ≥ 7 rated the feedback "specific and useful"; 100% of claims carry an evidence citation; 0 banned claims emitted.

Q3 · Months 7-9 (Feb-Apr 2027) — "Make It a Product"

Theme: app, scale, retention.

Milestone	Deliverable	Owner
M3.1 Platform decision	Pick ONE. Recommendation: mobile web (PWA) first — one codebase, no app-store review, instant iteration. Native later, driven by evidence.	Both
M3.2 App v1	Upload → capture-gate feedback → results → overlay playback synced to metrics	FT
M3.3 Backend + storage	FastAPI, S3, auth, consent capture, retention policy	FT
M3.4 Progress tracking	Session-over-session comparison — the #1 retention driver	PT
M3.5 Baseline library v2	30+ athletes per cohort; pro tier added	PT
M3.6 3D lifting (if deferred from Q1)	Restore as an accuracy upgrade on a working 2D product	FT
M3.7 Bat tracking	NOT DONE Cut from the 12-month plan. With the revised baseline there is no capacity for it. Year 2.	—
M3.8 Open beta	50+ athletes , self-serve	Both

Q3 exit gate: 50 athletes onboarded; $\geq 40\%$ return for a second session; cost/swing < \$0.15; p95 turnaround < 5 min.

Q4 · Months 10-12 (May-Jul 2027) — "Harden Batting, Spike Pitching"

Theme: ▲ **Revised.** The pitching *alpha* is cut; a pitching **technical spike** replaces it. With a recovery quarter absorbed at the start of the year, shipping a second sport would mean shipping both badly. Batting retention is the more valuable asset at month 12.

Milestone	Deliverable	Owner
M4.1 Batting hardening	Reliability, turnaround, cost, and retention work on the live product. This is the priority.	FT
M4.2 Pitching spike	9-event model + arm-slot classifier running on the shared backbone, validated offline against OpenBiomechanics pitching data. No users.	FT
M4.3 Safety policy	Banned-claim validator built and tested before any pitching output is ever shown	Both
M4.4 Pitching baselines	Overhand + three-quarter, HS tier, seeded from OpenBiomechanics	PT
M4.5 Reuse report	Quantified code reuse from the spike; expansion cost model for pitching, golf, tennis	Both
M4.6 Year-2 plan	Written on Year-1 evidence, not on the original vision doc	Both

Q4 exit gate: batting product stable with $\geq 40\%$ retention holding at scale; pitching spike validated offline; **$\geq 70\%$ infrastructure reuse demonstrated** (the extensibility claim proven without the cost of a second launch).

Why cut the pitching alpha rather than compress batting? A batting product 50 athletes use weekly is a real asset — for revenue, for recruiting, and for any funding conversation. A pitching alpha with 10 users and a shaky batting product is neither. The reuse report still proves the multi-sport thesis; it just proves it with evidence instead of a launch.

Timeline at a glance

	Q1 Aug-Oct	Q2 Nov-Jan	Q3 Feb-Apr	Q4 May-Jul
	RESTORE	TRUST	PRODUCT	HARDEN + SPIKE
FT	regression tests batter selection pose stabilize 3D lifting fix capture gate	event detection evidence packet agent services VLM fast lane validation report	app v1 backend 3D (if deferred) progress tracking	batting hardening pitching spike (offline, no users) safety validator
PT	golden set label capture protocol	baseline lib v1 alpha recruiting	baseline lib v2 beta recruiting	pitching baselines reuse analysis
USERS	0	10 (closed)	50+ (open beta)	50+ batting only
GATE	pipeline restored	MAE<15ms, 7/10	40% return	70% reuse proven

What changed from v1.0 and why: the revised baseline absorbs a full recovery quarter. Event detection slips Q1→Q2, the alpha slips Q2 (unchanged, but tighter), bat tracking is cut outright, and the pitching alpha becomes an offline spike. **Nothing was compressed to preserve the original dates** — that would only relocate the failure to a worse quarter.

9. Key Deliverables Summary

Documents: capture protocol · parental consent form · metric definitions (versioned) · validation report vs. OpenBiomechanics · banned-claim policy · Year-2 plan

Data: golden set (200–500 labeled swings) · baseline reference library v1/v2 · own-capture corpus (US + Korea) · OpenBiomechanics-derived validation set

Code: event detector · capture quality gate · Evidence Packet schema + validator · MEDA/APA/APE services · baseline comparison engine · app v1 · FastAPI backend · pitching event model + arm-slot classifier · (conditional) TrackNet bat tracker

Product: closed alpha (10) → open beta (50+) → pitching alpha (10)

10. Limitations & Constraints — Two People, Two Continents

10.1 The honest constraints

Constraint	Real consequence
~50 hrs/week combined	~1 major workstream + 1 support workstream. Not 4 sports, not 3 platforms.
13-14 hour time gap (US ↔ Korea)	Near-zero overlap hours. Anything requiring live debugging together will stall for a day per exchange.
Single points of failure	Two people means two bus factors of 1. Illness, a job change, or burnout stops the project outright.
No specialist depth	Neither of you is a full-time biomechanist, iOS engineer, or ML researcher. Some problems will take 3× longer than they would for a specialist.
No ground-truth hardware	No force plate, no marker mocap, no radar. You are dependent on public datasets and partnerships for validation.
Labeling is a hard floor	300 labeled swings ≈ 15–20 hours of human work that cannot be automated away or bought cheaply at this scale.

10.2 Turning the time gap into an advantage

The 13-hour gap is usually framed as pure cost. Structured correctly it is close to continuous progress — but only under strict discipline:

- **Split by ownership, never by task.** The full-time person owns the pipeline; the part-time person owns data, labeling, baselines, and user recruiting. Interfaces between them are files and schemas, not conversations.
- **Contracts before code.** Agree the JSON schema first, then both sides build against it independently. This is why the Evidence Packet matters organizationally as much as technically.
- **Async by default.** Written decision records (you already have MEMORY.md — keep it current), recorded walkthroughs instead of live demos, PR descriptions that assume the reader is asleep.
- **One weekly sync, non-negotiable.** 60 minutes, agenda written in advance, decisions logged. One real meeting beats five ad-hoc ones that only one person can attend.
- **Korea cohort as generalization test.** Two capture populations in two baseball cultures is a genuine technical asset. Most two-person teams cannot claim geographic diversity in their training data.

10.3 What to explicitly NOT do in the next 12 months

Scope discipline is the highest-value management act available to you. Written down so it can be pointed at when temptation arrives:

- **NOT DONE** Golf and tennis — architected for, not built
- **NOT DONE** Native iOS + Android + web simultaneously — one platform, chosen on evidence
- **NOT DONE** Exit velocity and bat speed — physics blocks them without calibration and ball tracking
- **NOT DONE** Injury-risk claims — no ground truth, and the users are minors
- **NOT DONE** Training pose models from scratch — fine-tune only on demonstrated systematic failure
- **NOT DONE** Real-time on-device inference — server-side until the pipeline is proven
- **NOT DONE** Multi-camera capture — kills the core "just a phone" value proposition

10.4 Growing the team — how to get help

Phase 1 (Q1-Q2): Advisors, not employees. Cheap, high leverage.

- **One biomechanics advisor** — a college or high-school strength coach, or a kinesiology grad student. Offer free analysis and co-authorship on any writeup. This is the highest-ROI relationship available: it validates your metric definitions and prevents you from confidently shipping something a domain expert would immediately reject.
- **2-3 coach design partners** — they tell you whether feedback is actually actionable, which no benchmark can.

Phase 2 (Q2-Q3): Build in public. Recruiting follows attention; attention follows artifacts.

- Publish the validation work — "here is our contact-detection error vs. marker-based mocap" is exactly the kind of post that reaches CV engineers who care.
- Open-source a genuinely useful, non-core piece: the capture quality gate, or Penn Action event-labeling tools. Signals competence, costs no moat.
- Present at a sports-analytics venue (SABR, SSAC). The baseball analytics community is unusually open and actively looks for this kind of work.
- **Korea advantage:** the Korean baseball market is large, underserved by English-language tools, and your partner is positioned there. A Korean-language product is real differentiation, not a translation task.

Phase 3 (Q3-Q4): Targeted contributors. Only after there is something worth joining.

Gap	Who	Engagement
Labeling throughput	Sports-science students	Paid hourly or course credit
Mobile/frontend	Contract RN or PWA dev	Fixed-scope, 4-6 weeks
Biomechanics validation	Kinesiology grad student	Research collaboration
Data partnerships	Facility owner	Data-for-analysis trade

What makes people join a two-person project: a working demo, a clearly scoped problem, and evidence you ship. You will have the first by Q2 and the third by Q3. Recruit then — not now, when the honest pitch is still "we have a pipeline and a plan."

Funding note: the Q2 alpha results and the OpenBiomechanics validation report are, together, a credible pre-seed narrative if you want one. Whether to raise is a Q3 decision informed by whether retention holds — not a Q1 distraction.

11. Risk Register

#	Risk	Likelihood	Impact	Mitigation
R0	Broken stages take longer than Q1 to restore	High	Critical	Timebox each fix (3 weeks for 3D). Take the 2D-only fallback rather than extending. The 2D overlay already works — it is a shippable floor.
R0b	Further silent regressions appear	Medium	High	M1.0 regression harness is the first deliverable of the year, specifically because this already happened once
R1	Event detection misses the < 15 ms target	Medium	Critical — blocks at-contact metrics	Audio fusion as a parallel path; if both fail, ship only non-contact metrics (separation, spine tilt, knee angle remain valid)
R2	Users shoot at 30 fps regardless of guidance	High	High	Capture gate + in-app camera with FPS enforcement; degrade gracefully to non-contact metrics rather than refusing service
R3	Labeling becomes the bottleneck	High	Medium	Start Week 1; active learning to prioritize informative frames; budget for paid student labelers in Q3
R4	One person becomes unavailable	Medium	Critical	Written decision records, no undocumented tribal knowledge, both sides able to run the full pipeline
R5	Metrics fail validation against OpenBiomechanics	Medium	High	Better to learn this in Q1 than post-launch. Narrow claims to what survives; that is a successful outcome, not a failure
R6	Bat tracking consumes the year	—	—	DONE Resolved by cutting it. Removed from the 12-month plan entirely. Body mechanics alone is a viable product.
R7	Scope creep toward golf/tennis	High	High	Section 10.3 exists to be pointed at
R8	Youth data/privacy misstep	Low	Critical	Consent forms from Q1; minimize retention; never train on non-consented footage

12. The Four Things That Matter Most

If everything else is noise, these are the load-bearing decisions:

1. **Restore the pipeline before extending it — and make it impossible to break silently again.** Three stages that were recorded as working are not. Build the regression harness first, then fix batter selection, pose, and 3D lifting in that order. Event detection is still the gate on every at-contact metric, but it cannot be built on a foundation that does not hold.
2. **Never let the model produce a number.** The deterministic/reasoning boundary is what makes this product defensible. Protect it in code, not in convention.
3. **Ship one sport, on one platform, to real users.** A batting product 50 athletes use weekly is worth vastly more than four half-built sports — as a business, as a technical asset, and as a recruiting story.
4. **Say what you cannot measure.** Physics blocks exit velocity, bat speed, and true 3D attack angle from a single uncalibrated phone. Competitors will overclaim these. Being the tool that is honest about uncertainty is a durable differentiator with coaches — the people who ultimately decide whether athletes adopt you.

Sources

- [The OpenBiomechanics Project](#)
- [driveline research/openbiomechanics — GitHub](#)
- [The OpenBiomechanics Project: Driveline Goes Open Source](#)
- [Analysis of Baseball Hitters at Different Levels of Competition — HAS-Motion](#)
- [AthleticsPose: Authentic Sports Motion Dataset](#)
- [Monocular 2D Baseball Swing Pose Estimation](#)
- Internal: `Baseball Big Picture.canvas`, `05-vision-local-llm-roadmap.md`, `CLAUDE.md`, `MEMORY.md`, `README.md`